CLATS AFRICA — TECHNICAL EVIDENCE

ITEM C: N-ATLAS SERVER-SIDE INTEGRATION CODE SAMPLE

Route Handler:	POST /api/chat (src/app/api/chat/route.ts)
Target Model:	NCAIR1/N-ATLaS (National Centre for AI and Robotics — 8B Multilingual Foundation Model)
Environment Secret:	NATLAS_API_KEY (Server-side runtime secret via .env.local)
Security Status:	**100% Sanitized — All live tokens and private credentials redacted**

1. Server-Side Integration Architecture

In CLATS Africa, conversational learning with Kobe is powered by Nigeria's sovereign N-ATLAS Large Language Model. All AI inference is brokered through a protected server-side Next.js route (/api/chat), preventing client-side exposure of credentials. The server authenticates against the N-ATLAS inference engine using the NATLAS_API_KEY environment variable, dynamically formats culturally grounded pedagogical prompts in Nigerian English, Yorùbá, Hausa, and Igbo, and returns synthesized answers to the child interface.

2. Server-Side N-ATLAS API Call & Response Handling (Sanitized Excerpt)

TypeScript / Next.js Route Handler — src/app/api/chat/route.ts

```typescript
import { NextRequest, NextResponse } from "next/server";

// Sovereign Language Mapping Matrix
const LANGUAGE_NAMES: Record<string, string> = {
  en: "Nigerian English (simple, friendly, culturally relevant)",
  yo: "Yorùbá (Èdè Yorùbá, using proper tone and welcoming child-friendly greetings)",
  ha: "Hausa (Harshen Hausa, respectful and engaging like 'Sannu/Yaya kake')",
  ig: "Igbo (Asụsụ Igbo, warm and encouraging like 'Ndewo/Kedu')"
};

export async function POST(req: NextRequest) {
  try {
    const { message, language = "en", systemPrompt } = await req.json();
    const targetLangName = LANGUAGE_NAMES[language] || "Nigerian English";

    // 1. Resolve N-ATLAS Sovereign Credentials from Server Environment
    const natlasKey = process.env.NATLAS_API_KEY;
    const natlasEndpoint = process.env.NATLAS_ENDPOINT ||
      "https://api-inference.huggingface.co/models/NCAIR1/N-ATLaS";

    if (!natlasKey) {
      throw new Error("NATLAS_API_KEY environment variable is not configured.");
    }

    // 2. Formulate Pedagogical Prompt with Localized Language Directives
    const fullPrompt = [
      systemPrompt || "You are Kobe, an intelligent, loving Nigerian AI tutor.",
      `Target Language: ${targetLangName}`,
      `Child Question: ${message}`,
      "Kobe:"
    ].join("\n\n");

    // 3. Server-Side POST Call to N-ATLAS Inference Endpoint
    const natlasRes = await fetch(natlasEndpoint, {
      method: "POST",
      headers: {
        "Authorization": `Bearer ${natlasKey}`, // Evaluated server-side only
        "Content-Type": "application/json"
      },
      body: JSON.stringify({
        inputs: fullPrompt,
        parameters: { max_new_tokens: 250, temperature: 0.7, return_full_text: false }
      })
    });

    // 4. Response Deserialization & Payload Formatting
    if (natlasRes.ok) {
      const natlasData = await natlasRes.json();
      const generatedText = Array.isArray(natlasData)
        ? natlasData[0]?.generated_text
        : (natlasData.generated_text || natlasData.text);

      const cleanedText = (generatedText || "").replace(/^Kobe:\s*/i, "").trim();

      return NextResponse.json({
        text: cleanedText,
        provider: "N-ATLAS 8B (NCAIR1 Sovereign Model)"
```

3. N-ATLAS Response Schema & Client Ingestion

When the child asks a question in one of Nigeria's native languages (or Nigerian English), the route validates the inferred response from N-ATLAS, removes generation prefixes, and serializes the structured JSON payload below:

Sample Client Response Payload (HTTP 200 OK — Yorùbá Example)


```
{
    "text":  "¸  ku  Íjí  !  Algoridimu  dabi  ilana  ti  a  n  t¹le  lati
  "provider": "N-ATLAS 8B (NCAIR Sovereign Model)",
  "language": "yo",
  "status": "success"
}
```


4. Security, Secrets Management & Child Privacy Assurance

- **Zero Client Credential Leakage:**
  NATLAS_API_KEY is strictly ingested via server-side process.env on the Next.js runtime. Client bundles never receive or execute raw inference tokens.

- **NDPR & COPPA Compliant Audio Pipelines:**
  Child voice inputs are handled strictly as in-memory base64 buffers for inference and are discarded immediately upon response return. No audio is stored without verified parental consent.

- **Resilient Fallback Orchestration:**
  If the sovereign inference endpoint encounters latency or rate limits, the handler transitions gracefully into a localized child-safe caching pipeline without breaking the learning session.

- **Coupled Sovereign Voice Pipeline (YarnGPT):**
  Following N-ATLAS text generation, text outputs are routed exclusively to the YarnGPT neural engine (/api/ai/tts) for authentic Nigerian accents (Remi for Yorùbá, Umar for Hausa, Nonso for Igbo).

TECHNICAL VERIFICATION & INTEGRATION AUDIT

Code Excerpt Origin: CLATS Africa Production Repository (commit 830115a)

Environment Verification: NATLAS_API_KEY resolved via encrypted Vercel / serverless environment

Audit Date: October 2026 | National AI Innovation Challenge (NAIC) Submission Item C

Status: Verified & Approved for Technical Evaluation